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Drill Problems 11

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Written by Jaehoon Song

1. **Solving for an Expression** (10 points)
Given that $2(x+4)=18$, what is the value of $x+6$?

   A. -4

   B. 5

   C. 9

   D. 11

   Answer:

   $\square$

2. **Inequality Solution** (10 points)

$$3x+2<-5(x+6)$$

   Which of the following values is a solution to the given inequality?

   A. -5

   B. -4

   C. 3

   D. 4

   Answer:

   $\square$

3. **Mean from Bar Graph** (10 points)

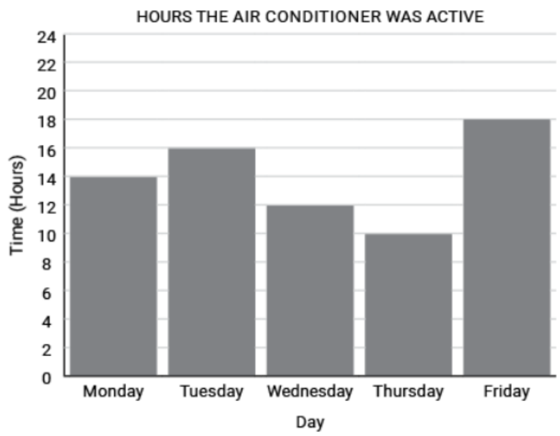

Figure 1: reference attached

The bar graph shows how many hours the air conditioner was on from Monday to Friday in a given week. What is the mean number of hours the air conditioner was on during this period?

A. 8
B. 10
C. 12
D. 14

Answer:

□

4. **Quadratic Expansion Constant** (10 points)
The expression $-2(x+3)^2+6$ is equivalent to ax^2+bx+c, where $a<0$ and $b<0$. What is the value of c?

Answer:

□

5. **Slope of Perpendicular Line** (10 points)
What is the value of the slope of a line perpendicular to the line $5x-6y+30=0$?

Answer:

□

6. **Sum of Quadratic Roots** (10 points)

$$3x^2 + 9x - 27 = 0$$

If m and n are solutions to the given equation, what is the value of $m + n$?

Answer:

□

7. **Point on a Circle** (10 points)
The graph of $(x-3)^2 + y^2 + 8y = 84$ is a circle in the xy-plane. Which of the following coordinates lie on the circle?

   A. $(1, 7)$
   B. $(-2, 5)$
   C. $(-3, 4)$
   D. $(3, -6)$

Answer:

□

8. **Stockpile Percentage Remaining** (10 points)
At the start of the winter, the US stockpile of corn was 5 billion bushels. Two months later, $p\%$ of the stockpile had been used. Which expression best represents the number, in billions, of bushels of corn remaining in the US stockpile at the end of those two months?

   A. $2(5)\left(\frac{100-p}{100}\right)$
   B. $5\left(\frac{100-p}{100}\right)$
   C. $(\frac{p}{100})$
   D. $2(5)(100-p)$

Answer:

□

9. **Profit Function Interpretation** (10 points)
The equation $P(y) = 37(1.1)^y$ gives the estimated annual profit, in thousands of dollars, at a pottery studio, where y is the number of years since the studio moved to a new location. Which of the following is the best interpretation of the number 37 in this context?

   A. The estimated annual profit, in thousands of dollars, when the studio moved to the new location
   B. The increase in the estimated annual profit, in thousands of dollars, each year
   C. The number of years since the studio moved to the new location
   D. The percent increase in the estimated annual profit each year

Answer:

□

10. **Isosceles Triangle Angle** (10 points)
Triangle JKL is an isosceles triangle. The measure of angle J is $30°$. If JL is the longest side of the triangle, what is the measure of angle K?

A. 30
B. 75
C. 120
D. 150

Answer:

□

11. **Function Value from Rule** (10 points)
If $f(2x) = 9x - 7$, what is the value of $f(6)$?

A. 11
B. 20
C. 47
D. 101

Answer:

□

12. **System of Equations Solutions** (10 points)

$$3x - 4y = 16$$
$$-6x = -8y + 32$$

How many solutions does the given system of equations have?

A. Exactly one
B. Exactly two
C. Infinitely many
D. Zero

Answer:

□

13. **Solving for Product** (10 points)
If $\frac{5}{x} = \frac{2}{y}$ and $\frac{4}{x} - \frac{2}{y} = -\frac{1}{5}$, what is the value of xy?

Answer:

□

14. **Radical and Exponent Equation** (10 points)
In the equation $\frac{1}{2}\sqrt{a} = 2\sqrt[3]{b}$, both a and b are positive real numbers. If $a = 8$ and the equation is written in the form $a^x = b$, what is the value of x?

Answer:

□

15. **Exponential Function from Points** (10 points)
The function g is defined by $g(x) = a^x - b$, where a and b are constants. In the xy-plane, the graph of $g(x)$ passes through the points $(0, -6)$ and $(1, 2)$. What is the value of a?

A. 6

B. 7

C. 8

D. 9

Answer:

□

16. **Quadratic Solution with Radical** (10 points)
If $1 + \frac{a\sqrt{2}}{2}$ is a solution to the equation $2x^2 - 4x - 7 = 0$ and $a > 0$, what is the value of a?

Answer:

□

17. **Transformation of Quadratic Function** (10 points)
If $f(x) = 2(x-3)^2 + 8$ is transformed into $g(x) = 2(x-5)^2 + 5$, which of the following describes the transformation?

A. The x-coordinate moves to the right 2 units and the y coordinate moves 3 units down.

B. The x-coordinate moves to the left 2 units and the y-coordinate moves 3 units down.

C. The x-coordinate moves to the right 2 units and the y coordinate moves 3 units up.

D. The x-coordinate moves to the left 2 units and the y-coordinate moves 3 units up.

Answer:

□

18. **Unit Conversion: Density** (10 points)
A sample of dried yellow pine has a density of 420 kilograms per cubic meter. To the nearest tenth, what is the density, in pounds per cubic foot, of this sample? (Use 1 kilogram = 2.2 pounds and 1 meter = 3.3 feet)

A. 5.3
B. 25.7
C. 127.3
D. 280.0

Answer:

□

19. **Arc Length in Circle** (10 points)

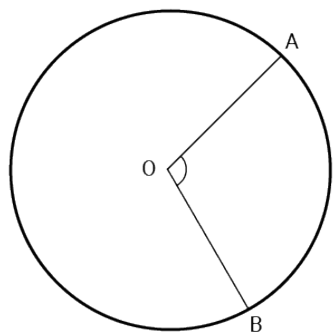

Figure 2: reference attached

In the given circle, the radius is 6 centimeters and angle AOB measures $\frac{7}{12}\pi$ radians. What is the value of the arc length AB, in centimeters?

A. $\frac{7}{2}\pi$
B. $\frac{21}{2}\pi$
C. 12π
D. 36π

Answer:

□

20. **Right Triangle Altitude Length** (10 points)

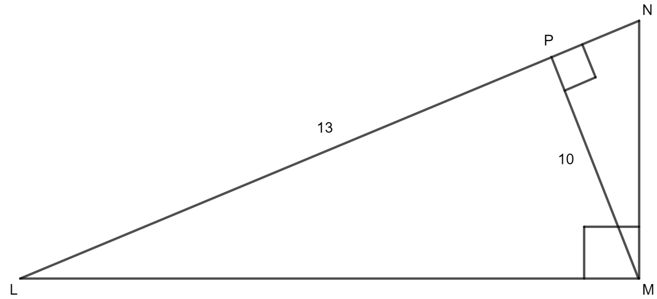

Figure 3: reference attached

In the image shown, right triangle LMN has a right angle at M. Altitude PM is shown and $\angle NPM$ is a right angle. $LP = 13$ and $PM = 10$. Which of the following are closest to the length of NP?

A. 3.0
B. 6.2
C. 7.1
D. 7.7

Answer:

□

21. **Quadratic from Table: Find** b (10 points)
$f(x) = 24x^2 + bx + c$
For the given function f, b and c are constants, and the graph of $y = f(x)$ in the xy-plane passes through the points in the table below. What is the value of b?

x	$f(x)$
$-\frac{15}{8}$	0
0	c
$\frac{11}{4}$	0

Answer:

□

22. **Polynomial Factors from Table** (10 points)

Select values for the polynomial function $g(x)$ are shown in the table. Based on the values in the table,

x	$g(x)$
-7	10
-4	0
0	5
3	1
5	-6
11	0

which of the following must be factors of $g(x)$?

A. $(x-5)$

B. $(x+5)$

C. $(x-4)$ and $(x+11)$

D. $(x+4)$ and $(x-11)$

Answer:

□